NL01CHEM01 Session 1 Problem Set

William Kang

3 November 2026

Edited and Formatted by Enoch Yu

These are six chemistry practice problems written by Vice President, William Kang. The problems are arranged roughly in increasing difficulty with subjective ratings, and covers the topic from William's presentation today. Enjoy!

§ 1 Problems

Problem 1. **Element, Compound, or Mixture? (• ○ ○)**
Write E, C, or M next to each.

a) Copper wire ( )
b) Table salt ( )
c) Well-stirred salt water ( )
d) Room air ( )
e) Distilled water ( )

Problem 2. **Pure, Uniform, or Non-Uniform? (• ○ ○)**
Label each as PURE, UNIFORM, or NON-UNIFORM.

a) Spoonful of stirred sugar water: _____
b) Handful of trail mix: _____
c) Sealed distilled water: _____
d) Rested Italian dressing: _____

Problem 3. **A Mystery Block (• • ○)**
A small solid block has mass 54 g and volume 20 cm^3. (a) Compute its density. (b) Which of the following elements is the solid block most likely made of: aluminum (2.7 g/cm^3), iron (7.9 g/cm^3), or copper (8.9 g/cm^3)?

Problem 4. **Counting Parts of an Atom (• • ○)**
An atom has atomic number 17 and mass number 35.

a) Which element is it?
b) Number of protons?
c) Number of neutrons?
d) Number of electrons if neutral?

Problem 5. **Average Atomic Mass (• • •)**
An element has two isotopes. Isotope M weighs 10.0 u and makes up 20.0% of atoms. Isotope N weighs 11.0 u and makes up 80.0%. Find the average atomic mass to one decimal place.

Problem 6. **Glass of Water (• • •)**
Use 18 g/mol for water and 6.02×10^{23} particles/mol.

a) How many moles in 180 g?
b) How many molecules total → write in scientific notation.
c) Roughly how many per sip if a glass is six sips?